

IMPLEMENTATION REPORT

AI & Automation Module

What we built, why we built it, and the value it delivers — covering AI usage tracking, automation rules, execution logging, and the RAG knowledge base powering the clinic assistant.

Backend framework	FastAPI
Modules covered	5 (usage tracking, rules, execution logs, knowledge docs, embeddings)
Vector storage	pgvector
AI provider	OpenAI

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1. What We Built — Modules & Endpoints

Inside the FastAPI application, five AI & Automation modules were designed, implemented, and tested end-to-end. Together they form the backend foundation for intelligent, cost-aware automation across the CRM.

1. AI Usage Tracking

GET/POST /ai/usage

Records how many tokens are consumed and how much it costs every time an AI model (e.g. OpenAI) is called. This gives a running, queryable log of AI spend across the whole application.

2. Automation Rules

GET/POST /ai/automation-rules

Lets you define rules of the form "when this event happens, run this AI action" — for example, when a new lead is created, automatically trigger an AI-generated summary.

3. Automation Execution Logging

GET/POST /ai/automation-runs

Every time an automation rule actually runs, its outcome (success/failure) and full execution details are logged to the database — giving a complete, auditable history.

4. Knowledge Documents

GET/POST /ai/knowledge/documents

Registers metadata for clinic FAQs, policies, and other reference documents — the foundation layer for Retrieval-Augmented Generation (RAG).

5. Knowledge Chunks & Embeddings

GET/POST /ai/knowledge/chunks

Splits document text into smaller chunks, generates vector embeddings for each chunk, and stores them in the database using pgvector — enabling fast semantic search.

Module summary

#	Module	Endpoint	Core function
1	AI Usage Tracking	/ai/usage	Token & cost logging
2	Automation Rules	/ai/automation-rules	Event → AI action definitions
3	Automation Execution Logging	/ai/automation-runs	Run history & status
4	Knowledge Documents	/ai/knowledge/documents	RAG source registry

#	Module	Endpoint	Core function
5	Knowledge Chunks & Embeddings	/ai/knowledge/chunks	Chunking + vector storage

2. Why We Built This — Core Purpose



Smart Automation

Move the clinic management system away from manual, repetitive work — automation rules let routine tasks run themselves the moment a triggering event happens.



AI Integration

Integrate OpenAI and vector search (RAG) directly into the system, so the AI layer actually understands clinic-specific data and can give accurate, grounded answers instead of generic ones.



Resource & Cost Control

Track exactly how much each AI feature costs to run, so usage can be monitored and the budget stays predictable and manageable.

3. Technical & System Benefits

Benefit	What it means in practice
Scalability	The system is now fully scalable — new automation rules and workflows can be added without any structural rework.
Data Integrity	Foreign key constraints and properly designed relational tables keep all data (rules, runs, chunks) secure, consistent, and organized.
Fast Retrieval	The pgvector-based vector database lets the AI find relevant information from documents in milliseconds, rather than scanning raw text.

4. What This Means for Users

Benefits across the user roles this module serves — Admin, Staff, and Patients:

Time Saving

Staff no longer need to manually write a summary for every lead — the moment a lead is created, the system automatically generates an AI summary in the background.

Better Customer / Lead Engagement

Automatic follow-ups and instant AI-driven responses mean leads don't slip through the cracks — which directly improves conversion rates.

Instant, Accurate Information

When patients ask about clinic timings, policies, or treatments, the AI knowledge base (built from the chunked, embedded documents) retrieves an instant, accurate answer grounded in the clinic's real content — not a generic guess.

Bottom line: these five modules turn the AI layer from a black box into a trackable, auditable, cost-controlled system — every AI action has a rule behind it, a log after it, and real clinic knowledge grounding its answers.